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The Hospital Ethics Challenge Simulation Game

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Medical Dilemmas: The Hospital Ethics Challenge

Introduction

In the modern world, the healthcare sector frequently faces difficult moral conundrums. Many times, doctors must make tough choices that could mean the difference between life and death. The "Medical Dilemmas: The Hospital Ethics Challenge" simulation game places players in the roles of a hospital administrator and a senior physician. Players must oversee staff management, patient care, resource allocation, and public perception in order to adhere to ethical standards and ensure the greatest possible patient outcomes.

This game not only provides an entertaining gaming experience but also serves as a teaching tool to analyze the ethical decisions made in the medical field.

Game Concept

With limited resources and a variety of medical problems, players assume control of a huge city hospital in "Medical Dilemmas" during a crisis. Players must make crucial decisions about who receives care, how resources are distributed, and how to maintain the hospital's operations and reputation. Every choice has an impact on patient outcomes, hospital reputation, and staff morale.

The game's primary goal is to solve moral dilemmas under pressure, producing an engaging and lifelike simulation that highlights the importance of moral judgment in the medical industry.

Objectives

- Create a simulation game that exposes players to moral and medical conundrums from real life.
- Give players a chance to investigate several ethical frameworks in a medical setting, such as utilitarianism, deontology, virtue ethics, etc.
- Compile and examine player choices to learn how various moral precepts affect medical judgment.
- Make use of the game as a teaching tool to promote conversations about healthcare ethics, including patient prioritization, end-of-life care, and resource allocation.

Gameplay Overview

The game begins with the user taking the role of a senior hospital administrator who is tasked with managing operations of a hospital in a crisis. He has to manage the supply of limited consumables like hospital beds, medications, as well as staff. Players are constantly given new challenges to deal with, such as critically ill patients, hospitals running low on supplies, and employees squabbling with one another.

Key Features:

1. Ethical Dilemmas:

- Players are regularly presented with tough ethical decisions, such as:
 - Who ought to get life-saving treatment in the scenario of scarce resources?
 - Is it justifiable to administer experimental treatment if there is no standard treatment available?
 - How is decision-making concerning end-of-life care handled when there are disagreements between the patient and the family?

2. Resource Management:

- Players have to allocate hospital resources like medical items, staff, and money. They need to fulfil the demands of their patients while also ensuring that their staff, resources, and money are not exhausted.

3. Moral Decision System:

- Every decision is assessed according to a different ethical system (utilitarian, deontological, etc.) and the participants are able to evaluate the level of their compliance with these frameworks.
- The game analyzes the player's decisions and provide them with feedback regarding their moral logic.

4. Realistic Medical Cases:

- The game consists of various medical cases including simple injuries and intricate critical care cases. For every case, the players must decide about the course of action to be taken to provide the care and the level of resources that ought to be supplied.

Core Game Scenarios

1. Viral Outbreak Scenario: There is a viral outbreak that overflows the hospital, and you have more patients than available beds. You must decide how to allocate the scarce resources—who gets treated first, and who must wait? You are faced with:

- A seriously ill child.
- A surgeon who can save more lives if he survives.
- A pregnant woman and unborn child, both at risk.
- Some patients who are unlikely to live even if treated.

All the options have implications for the potential of the hospital to save lives, and you must weigh the value of saving more than one patient against resources.

2. Organ Donation Dilemma: Both patients need an organ transplant, but there is only a single organ. One patient is a young working individual with a promising future, and the other is middle-aged with dependents. The decision is yours whom will receive the organ?

3. Care at the End of Life: A terminally ill old patient requests to be taken off life support. The patient's family, however, insists that all that can be done should be done to prolong life. The hospital staff holds off waiting for you to make a decision. Do you honor the patient's request or abide by the insistence of the family?

4. **Medical Research Ethics:** An experimental drug is provided by a pharmaceutical company to treat a dying patient. The treatment is not proven but provides hope for recovery. You have to make a decision on whether to let the patient receive the experimental treatment, even though there are risks involved.

Tools & Technologies

Frontend Development:

- **Unreal Engine** will be utilized to create a realistic and immersive hospital environment. The game's advanced graphical capabilities will ensure a lifelike simulation, enhancing player engagement with the ethical dilemmas presented.

Backend Development:

- **Flask**, a lightweight Python framework, will manage game logic, API requests, and data processing. Its simplicity makes it ideal for handling the game's decision-making mechanics.

Database:

- **MySQL** will store player profiles, medical cases, decisions, and outcomes, allowing for detailed analysis of player choices and ensuring data integrity.

UI/UX Design:

- **Sketch** will be used to design a clean, minimalistic user interface, keeping players focused on the ethical dilemmas without unnecessary distractions.

Expected Outcomes

- **Engaging Gameplay:** A compelling, immersive simulation that challenges players to make difficult ethical decisions in a high-stakes medical environment.
- **Educational Value:** The game will serve as a platform for teaching ethics and morality, fostering discussion about complex medical dilemmas and real-world scenarios.
- **Data Collection:** The game will provide insights into how people from different backgrounds and cultures approach medical ethics. This data could be used for academic research or public policy discussions.
- **Moral Reflection:** Players will receive feedback on their decisions, allowing them to reflect on their moral reasoning and understand how their choices align with ethical frameworks.

Conclusion

"Medical Dilemmas: The Hospital Ethics Challenge" is a game and simulation, but it is also an educational and challenging experience that gives the importance of ethics in health care its due. Through examining how players handle challenging moral dilemmas, the game delivers sound lessons on the human decision-making process and highlights the practical importance of ethical thinking in medicine.